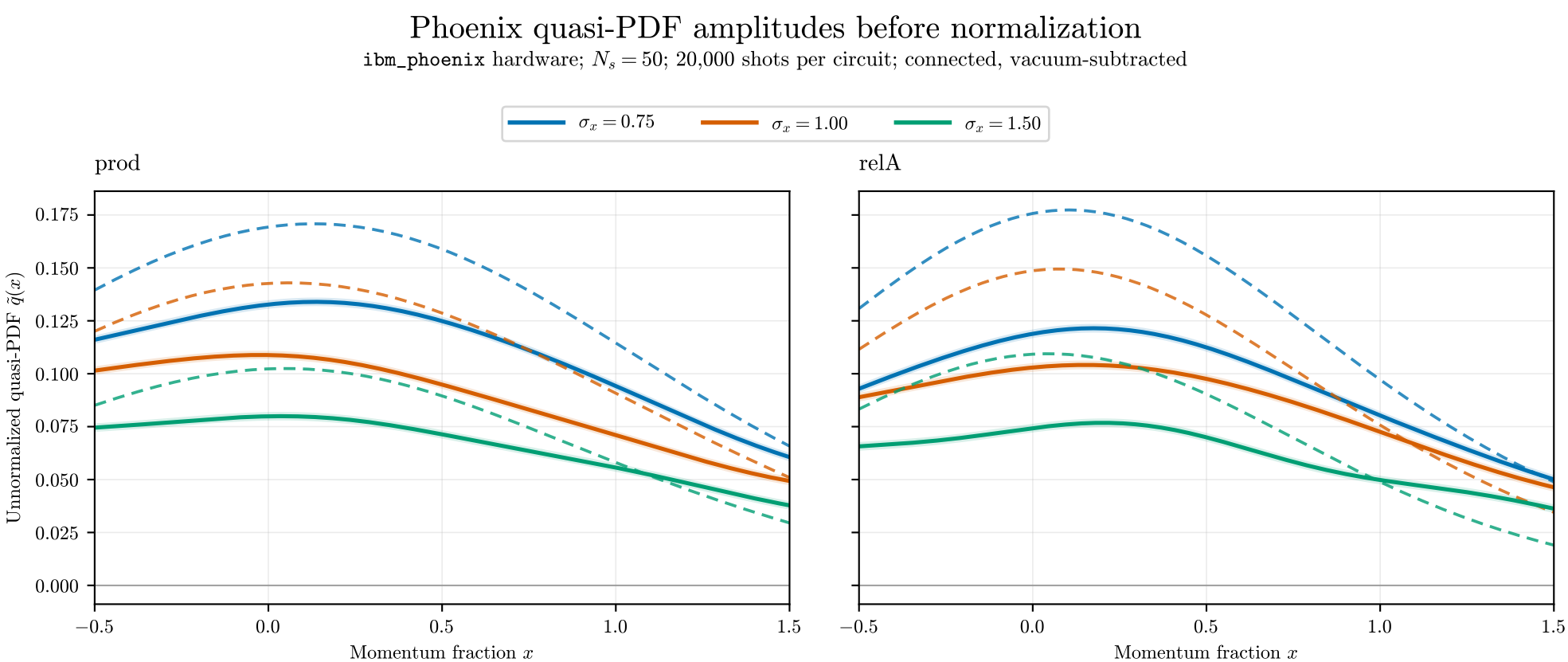

Phoenix quasi-PDF amplitudes before normalization
ibm_phoenix hardware; $N_s = 50$; 20,000 shots per circuit; connected, vacuum-subtracted
$\sigma_x = 0.75$
$\sigma_x = 1.00$
$\sigma_x = 1.50$
prod
relA
Unnormalized quasi-PDF $\tilde{q}(x)$
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
−0.5
0.0
0.5
1.0
1.5
Momentum fraction x
Solid and bands: hardware with shot errors; dashed: ideal MPS reference
Vacuum-subtracted; $|m| \leq 5$, Gaussian window $\sigma_m = 5$. Bands: 1σ, shot noise only.